OMI---IMO Canonical Resolution

Chiral URI-Compatible Place-Value Notation for Pseudo-Persistent Porting

Status: Canonical Resolution Draft
Scope: OMI notation, OMI-Port, OMI-Lisp declaration surfaces, URI-CIDR scope notation, chiral frame rendering
Supersedes: Debate framing of `omi---imo` / `o---o/---/?---?@---@`
Preserves as historical: LL/Ket 0xE_ manifold model
Canonical authority: FS/GS/RS/US pre-language memory topology

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0. Resolution

The canonical compact OMI route form is:

```
omi---imo
```

The canonical expanded OMI-IMO relation surface is:

```
o---o/---/?---?@---@
```

The canonical machine expansion is:

```
o-FS-GS-RS-US/FS/GS/RS/US?REGISTER?STACK@CAR@CDR
```

The canonical CONS spine is:

```
(omi---imo . (o---o . (/---/ . (?---? . @---@))))
```

This notation is now resolved as:

```
a URI-compatible, chiral, mixed-radix, place-value memory ruler
for pseudo-persistent porting annotations over sign-value data.
```

It is not merely a string.
It is not merely a URI.

It is not merely a Lisp expression.
It is not a runtime connection.
It is not validation.
It is not receipt.

It is the canonical printable address-ruler surface for OMI-IMO candidate relations.

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1. Authority Boundary

OMI begins as scoped memory topology, not as a function call.

The older `LL/MM/NN` model is deprecated as canonical authority. The canonical model is:

```
FS / GS / RS / US
```

with the pre-language control alphabet:

```
0x00  NUL  origin / null / zero-point
0x1C  FS   file scope
0x1D  GS   group scope
0x1E  RS   record scope / closure witness
0x1F  US   unit scope
0x20  SP   readable boundary / visible separator
```

The canonical rule is:

```
place first
relation second
interpretation third
validation fourth
receipt last
```

The uploaded FS/GS/RS/US authority draft states that the old `LL/MM/NN` model is replaced by FS/GS/RS/US as pre-language memory topology and explicitly defines `omi---imo`, `o---o/---/?---?@---@`, and the expanded machine form with FS/GS/RS/US plus REGISTER/STACK/CAR/CDR.

Therefore:

```
control is not acceptance
scope is not acceptance
relation is not acceptance
```

```
closure is not acceptance
projection is not acceptance
validation and receipt accept
```

---

2. Canonical Frame

The OMI-IMO frame is a 256-bit memory relation.

```
full frame:
    256 bits
    32 bytes
    64 nibbles
```

It has two halves:

```
ruler half:
    128 bits
    16 bytes
    32 nibbles
    S0 S1 S2 S3 S4 S5 S6 S7

payload / pair half:
    128 bits
    16 bytes
    32 nibbles
    REGISTER STACK CAR CDR
```

The ruler half is:

```
S0-S1-S2-S3/S4/S5/S6/S7
```

with canonical meaning:

```
S0 = FS
S1 = GS
S2 = RS
S3 = US

S4 = FS'
S5 = GS'
```

```
S6 = RS'
S7 = US'
```

The payload/pair half is:

```
?REGISTER?STACK@CAR@CDR
```

The complete expanded form is:

```
o-FS-GS-RS-US/FS/GS/RS/US?REGISTER?STACK@CAR@CDR
```

The source authority document defines the same 256-bit split: first 128 bits as the scoped ruler cascade and second 128 bits as the REGISTER/STACK/CAR/CDR payload-pair cascade.

---

3. Separator Classes

The canonical separator classes are:

```
-   OMI forward scope / citation route
/   IMO inverse scope / carrier route
?   payload / register / stack / mask / witness query field
@   CAR/CDR relation pointer
.   OMI-Lisp pair constructor after SP
```

Thus:

```
o---o
```

is the forward OMI route surface.

```
/---/
```

is the inverse/carrier IMO route surface.

```
?---?
```

is the witness/query/register/stack surface.

```
@---@
```

is the CAR/CDR pointer surface.

Together:

```
o---o/---/?---?@---@
```

is the expanded URI-compatible OMI-IMO relation surface.

---

4. URI Compatibility

OMI notation is URI-compatible, not URI-dependent.

The notation intentionally uses characters that can survive common URI and text surfaces:

```
-
/
?
@
.
```

But these characters are interpreted under OMI memory topology, not ordinary web semantics.

A web URI may be used as an OMI-Port scope. A URI-CIDR form may be used as a default printable scope grammar. But web URL authority is not OMI authority.

Canonical rule:

```
URI names.
CIDR scopes.
OMI frames.
Validation accepts.
Receipt records.
```

---

5. ASCII Omicron Chirality

OMI may render omicron frame delimiters using ASCII `O` and `o`.

```
'O' = 0x4F
'o' = 0x6F

0x6F - 0x4F = 0x20
```

The same byte value:

```
0x20
```

is `SP`, the readable boundary.

Therefore ASCII provides a direct algorithmic chirality cue:

```
chirality_bit = byte & 0x20;
```

or:

```
flipped = byte ^ 0x20;
```

Canonical chiral forms:

```
o---O
    forward low-to-high chiral frame

O---o
    inverse high-to-low chiral frame

o---o
    neutral canonical carrier fallback

O---O
    high-high sealed rendition
```

Resolution:

```
O/o carries chirality.
--- carries the URL-compatible stuffed route span.
```

The hyphen is not itself the `0x20` bit. The hyphen-run is the visible rail governed by the endpoint chirality.

---

6. Unicode Omicron and Bidi Projection

Unicode Greek omicron may render the same relation:

```
o---O
```

But Unicode is a rendition layer.

Canonical carrier authority remains ASCII unless a specific binary encoding declares otherwise.

```
ASCII O/o:
    carrier chirality

Greek o/O:
    high-rendition display

Unicode bidi:
    projection / inverse traversal display
```

Unicode bidi may be used to display the inverse route direction at no change to the canonical logical byte order.

Canonical rule:

```
bidi may project
bidi may mirror
bidi may aid inspection
bidi must not alter canonical bytes
bidi must not validate
bidi must not receipt
```

Thus the parser must operate on logical byte order. Renderers may show left-to-right or right-to-left projections.

---

7. Dot Is Earned After SP

Dot notation is not primitive at byte zero.

The readable boundary is:

```
0x20 SP
```

Only after SP may OMI-Lisp express pair declarations such as:

```
(car . cdr)
```

Therefore:

```
(omi---imo . (o---o . (/---/ . (?---? . @---@))))
```

is a readable declaration surface, not a pre-language byte-zero primitive.

Canonical order:

```
pre-language scope
→ SP readable boundary
→ dot relation
→ candidate
→ validation
→ receipt
```

This preserves the OMI-Lisp rule:

```
OMI-Lisp declares.
OMI-Lisp does not accept.
```

---

8. CONS Spine Resolution

The canonical expanded notation may be read as a CONS spine:

```
(omi---imo . (o---o . (/---/ . (?---? . @---@))))
```

This means:

```
omi---imo
    compact reversible frame
```

```
o---o
    forward scope surface

/---/
    inverse carrier surface

?---?
    query / witness / register / stack surface

@---@
    CAR/CDR relation surface
```

The same object can be traversed in two directions:

```
left to right:
    declare → scope → carrier → witness → relation

right to left:
    relation → witness → carrier → scope → declaration
```

This is why the notation behaves like a slide rule and a reversible address ruler.

---

9. Mixed-Radix Place-Value Resolution

The OMI-IMO frame is place-value notation.

Canonical scale facts:

```
1 nibble  = 4 bits  = 16 values
4 nibbles = 16 bits = one 0x0000 field
8 nibbles = 32 bits = one 0x00000000 field

32 nibbles = 128 bits
64 nibbles = 256 bits
```

The notation supports multiple radix interpretations:

```
base16:
    hex nibble address surface

base4:
    quadrant / two-bit subdivision surface
```

```
base2:
    chirality / polarity / bit-mask surface

base60:
    sexagesimal angular / time / divisor coordination analogy
```

A fully qualified address can be interpreted as nested base-radix monoids:

```
(omi-base16-base16-imo
  . (o-(base4 . base4)-(base4 . base4)-o
  . (/-(base4 . base4)-(base4 . base4)-/
  . (?-(base4 . base4)-(base4 . base4)-?
  . @-(base4 . base4)-(base4 . base4)-@))))
```

A chiral variant may use endpoint case:

```
(o_base4 . O_base4)
```

to indicate low-to-high ASCII chirality within a base4 subdivision.

---

10. Slide Rule and Nomogram Interpretation

`|OMI---IMO>` is a slide-rule-like annotation.

A slide rule does not own multiplication. It aligns scales.

Likewise, OMI notation does not own data. It aligns place-value surfaces.

Canonical interpretation:

```
|OMI---IMO>
    start / stop frame

o---o
    forward ruler

/---/
    inverse carrier ruler

?---?
    witness / query / register ruler
```

```
@---@
    CAR/CDR relation ruler
```

Thus `|OMI---IMO>` gives sign-value data:

```
start point
stop point
scope
direction
subpath
relation surface
```

This makes it a pseudo-persistent porting annotation.

---

11. Pseudo-Persistent Porting Annotation

A pseudo-persistent porting annotation is:

```
a stable notation wrapper that gives sign-value data a portable address frame
without claiming to own, validate, mutate, connect, or accept the data.
```

Therefore:

```
|OMI---IMO>
```

may annotate:

```
text spans
binary blobs
DOM nodes
files
URIs
LoRa frames
sensor events
packets
receipts
candidate relations
```

The annotation says:

```
this sign-value is placed on an OMI ruler
and may be ported through OMI place-values
but it is not accepted state yet
```

Canonical rule:

```
annotation is not authority
porting is not connection
projection is not acceptance
validation and receipt accept
```

---

12. Gnomic Omicron

The omicron marker is gnomic.

A gnomon is a reference marker that lets displacement, angle, shadow, or growth become measurable.

In OMI:

```
O/o
    gives frame orientation

---
    gives the span

FS/GS/RS/US
    gives scoped place-value topology

REGISTER/STACK/CAR/CDR
    gives relation payload fields
```

Thus the gnomic omicron is:

```
the visible frame marker that lets a resolver measure how sign-value data moves
through place-value space.
```

The four-dimensional analogy is permitted as explanatory geometry:

```
3D cube displacement:
    d² = x² + y² + z²
```

```
4D tesseract displacement:
    d² = w² + x² + y² + z²
```

In three dimensions, Legendre's three-square theorem leaves excluded square-root distances. In four dimensions, Lagrange's four-square theorem allows every natural number to appear as a sum of four squares. This supports the interpretation of 4D unfolding as a fuller distance alphabet.

This is explanatory geometry, not protocol authority.

---

13. F* Gauge Family Resolution

The OMI gauge pre-header is:

```
G 00 1C 1D 1E 1F 20 G
```

with:

```
G ∈ 0xF0..0xFF
```

The canonical default is:

```
FF 00 1C 1D 1E 1F 20 FF
```

The F* family may encode the Wittgenstein truth-function row in the low nibble:

```
F0 = contradiction
F1 = NOR
F2 = converse nonimplication
F3 = not-p
F4 = material nonimplication
F5 = not-q
F6 = XOR
F7 = NAND
F8 = AND
F9 = XNOR
FA = q
FB = p → q
FC = p
FD = p ← q
```

```
FE = OR
FF = tautology / canonical closure
```

Resolution:

```
F0..FF defines a gauge dialect family.
The low nibble selects the truth row.
FF seals tautological canonical closure.
```

This is compatible with the FS/GS/RS/US pre-language cascade because the pre-header embeds:

```
00 1C 1D 1E 1F 20
```

inside the gauge boundary.

---

14. Deprecated LL/Ket Material

The LL/Ket document is preserved as historical and explanatory, not canonical authority.

It contributed useful phrases:

```
The high nibble carries.
The low nibble selects.
The first twelve pack.
The last four hold.
```

It also defined the old rule:

```
slot_index = LL & 0x0F
rule_slot  = 0xE0 + slot_index
context    = (LL >> 4) & 0x0F
```

The LL/Ket document describes `LL` as a ket-like scope selector and separates the `0xE0-0xEB` inner materializer field from the `0xEC-0xEF` outer controller vertices.

Resolution:

```
LL/Ket is deprecated as root authority.
Its nibble intuition may remain as historical materializer explanation.
FS/GS/RS/US is canonical memory topology.
```

---

15. OMI-Port Consequence

In `omi-port`, `|OMI---IMO>` becomes the default dormant porting ruler.

A port is not a connection.

A port is:

```
a dormant typed opening over an OMI place-value scope
```

A binding is:

```
a declared source-target displacement
```

A path is:

```
a composed ruler traversal
```

A pipe is:

```
an authorized active stream, outside omi-port v0
```

Canonical port doctrine:

```
OMI-Port declares a calibrated source-target displacement over an OMI place-
value ruler.

It does not connect.
It does not send.
It does not receive.
It does not mount.
It does not mutate.
```

```
It does not validate.
It does not receipt.
```

---

16. Canonical Implementation Rules

A conforming parser or resolver MUST observe:

```
1. Parse logical byte order, not visual bidi order.

2. Treat `omi---imo` as the compact route form.

3. Treat `o---o/---/?---?@---@` as the expanded relation surface.

4. Treat `o-FS-GS-RS-US/FS/GS/RS/US?REGISTER?STACK@CAR@CDR`
   as the machine expansion.

5. Treat ASCII O/o as optional chirality annotation.

6. Treat Greek ο/Ο as rendition, not carrier authority.

7. Treat dot notation as post-SP readable relation syntax.

8. Emit candidates with:
       accepted = 0
       validated = 0
       receipted = 0

9. Require downstream validation and receipt for accepted state.
```

---

17. Canonical Summary

```
Compact:
    omi---imo

Expanded:
    o---o/---/?---?@---@

Machine:
    o-FS-GS-RS-US/FS/GS/RS/US?REGISTER?STACK@CAR@CDR

CONS:
```

```
    (omi---imo . (o---o . (/---/ . (?---? . @---@))))

Binary:
    256 bits
    32 bytes
    64 nibbles

Ruler:
    first 128 bits = FS/GS/RS/US cascade
    second 128 bits = REGISTER/STACK/CAR/CDR relation surface

Chirality:
    O/o differ by 0x20
    chirality_bit = byte & 0x20

Projection:
    Unicode may render
    bidi may invert display
    logical byte order remains canonical

Authority:
    declaration is not acceptance
    porting is not connection
    validation and receipt accept
```

---

18. Final Canon

`OMI---IMO` is the canonical compact reversible route notation.

`o---o/---/?---?@---@` is the canonical expanded URI-compatible OMI-IMO relation surface.

The notation is a chiral, mixed-radix, place-value memory ruler. It can be parsed as a CONS spine, rendered as a URI-compatible address surface, projected through Unicode bidi, and interpreted as a slide-rule or nomogram for sign-value data.

ASCII `O/o` supplies an algorithmic chirality cue through the `0x20` case bit. The hyphen-run supplies the URL-compatible stuffed route span. FS/GS/RS/US supplies canonical pre-language memory topology. REGISTER/STACK/CAR/CDR supplies the relation surface.

Therefore:

```
|OMI---IMO>
```

is resolved as:

```
a pseudo-persistent porting annotation for sign-value data
over a reversible OMI place-value ruler.
```

It declares where a relation may land.
It does not connect.
It does not validate.
It does not receipt.
It does not accept.

Validation and receipt remain the only acceptance boundary.

One-line canon:

```
OMI---IMO is the chiral URI-compatible slide-rule notation for scoped sign-value
porting: a pseudo-persistent address ruler whose visible CONS spine lowers into
FS/GS/RS/US memory topology and remains candidate-only until validation and
receipt.
```